

TANVI WAMORKAR

Postdoctoral Research Associate | Stanford University

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EMPLOYMENT

2025–Present	Postdoctoral Research Associate, Stanford University
2024–2025	Postdoctoral Research Associate, University of California, Berkeley
2021–2024	Postdoctoral Research Associate, Argonne National Laboratory

EDUCATION

2021	Ph.D. Physics, Northeastern University
2014	M.Sc. Physics, University of Delhi
2012	B.Sc. Physics, University of Delhi

RESEARCH EXPERIENCE

2024–present	ML-based Unfolding & Likelihood Ratio Estimation Methods <i>ATLAS Run 2 Z+Jets Unbinned Measurement</i> • Explored Lorentz-equivariant network architectures to optimize neural network performance • Developed a neural network-based strategy to estimate and subtract top-quark background, improving accuracy in high-background regions of phase space <i>H1</i> • Contributed to framework development for first unbinned measurement of entire phase space • Developing public-facing package to publish neural networks to accompany the publication <i>Software & Method Development</i> • Exploring the impact of explicit versus implicit physics priors on likelihood ratio estimation and anomaly detection • Developed ensemble strategies for neural networks to minimize bias and variance in likelihood ratio estimation • Developed a RooUnfold implementation of Omnifold and released it as an accessible Python package for unfolding tasks
2021–present	Standard Model Measurement <i>ATLAS Run 3 WWW Cross-Section Measurement</i> • Analysis contact for a working group of 20 physicists from 5 institutes • Developed lightweight analysis framework for processing and analyzing data • Developing multiclassifier deep neural network to improve signal sensitivity and to perform Effective Field Theory interpretations
2018–2021	Higgs Physics <i>CMS Higgs Exotic Decay Search</i> • Analysis contact and primary editor of analysis note and journal publication • Developed neural network-based photon reconstruction algorithm with 10% performance improvement vs. CMS default • Designed an end-to-end deep learning technique to estimate the invariant mass of merged photon candidates
2018–2024	Detector Upgrade for High Luminosity-LHC <i>ATLAS Inner Tracker (ITk) Upgrade (2021-2024)</i> • Led ITk pixel module assembly, testing, and cleanroom readiness at Argonne, including DAQ and coldbox operations • Developed automation software that improved module metrology and streamlined production workflows • Contributed to collaboration-wide quality control and design efforts, including site qualification reviews across 15 institutes and documentation for Inner System components <i>CMS Endcap Timing Layer Upgrade (2018-2020)</i>

- Built a cold box at Fermilab to characterize prototype sensors and front-end electronics; developed software for efficient test-beam data taking
- Led test-beam efforts at Fermilab measuring LGAD sensor hit efficiency and time resolution

2016–2021

Detector Operations

CMS ECAL Alignment (2016-2019)

- Performed ECAL–tracker alignment; developed a Python tool used in the CMS control room to monitor alignment status

CMS ECAL DAQ (2016-2018)

- Served as on-call ECAL data-acquisition expert throughout LHC Run 2 operations
- Performed CMS ECAL detector on-call and prompt-feedback shifts
- Designed and deployed a Node.js-based payload monitor for ECAL electronics, contributing to a >6% reduction in luminosity loss

SELECTED PUBLICATIONS

This section contains publications in which I was a primary analyzer or made significant contributions. A full list of my publications can be found on my *InspireHEP* profile.

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| 2025 | H1 Collaboration. [Preliminary] Towards Unfolding All Particles in High Q^2 DIS Events . H1PRELIM-25-031 |
| 2025 | F.T. Acosta, T. Wamorkar, V. Mikuni, B. Nachman. Stabilizing Neural Likelihood Ratio Estimation . ARXIV:2503.20753 |
| 2025 | R. Milton, V. Mikuni, T. Lee, T. Wamorkar, B. Nachman. Tools for unbinned unfolding . JINST 20 (2025) 05, P05034 |
| 2023 | CMS Collaboration. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons . JHEP 07 (2023) 148 |
| 2023 | CMS Collaboration. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$. PHYS. REV. LETT. 131, 101801 |
| 2022 | CMS Collaboration. The CMS MTD Endcap Timing Layer: Precision timing with LGADs . NIMA 1032 (2022) 166627 |
| 2021 | CMS MTD Collaboration. Test beam characterization of sensor prototypes for the CMS Barrel MIP Timing Detector . JINST 16 (2021) P07023 |
| 2021 | CMS Collaboration. Electron and photon reconstruction and identification in CMS . JINST 16 (2021) P05014 |
| 2021 | Heller et al. Combined analysis of HPK 3.1 LGADs for high-volume QC . NIMA 1018 (2021) 165828 |
| 2020 | CMS Collaboration. Reconstruction of ECAL signal amplitudes under pileup . JINST 15 (2020) P10002 |
| 2019 | CMS Collaboration. A MIP Timing Detector for the CMS Phase-2 Upgrade . CERN-LHCC-2019-003 |
| 2019 | T. Wamorkar. CMS ECAL calibration and alignment . NIMA 936 (2019) 121–123 |
| 2019 | T. Wamorkar. Performance of the CMS ECAL data acquisition system in Run 2 . NIMA 936 (2019) 346–348 |

CONFERENCE TALKS AND POSTERS

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| 2025 | <i>Advancing Particle Physics Measurements with Machine Learning [Poster]</i>
Center for Decoding the Universe Quarterly Forum |
| 2025 | <i>Explicit versus implicit physics priors for separating nearly identical classes</i>
ML4Jets |
| 2023 | <i>Measurement of electroweak gauge boson production in association with jets in ATLAS</i>
QCD@LHC 2023 |
| 2023 | <i>Module assembly and testing for the ATLAS Phase-II Inner Tracker Pixel Detector</i>
APS April Meeting |
| 2023 | <i>Module Metrology</i>
US ITk Pixels Workshop |
| 2021 | <i>Exotic Higgs decays at CMS</i>
Higgs 2021 |
| 2021 | <i>Search for exotic decays of the Higgs Boson using photons with the CMS experiment</i>
European Physical Society Conference on High Energy Physics |

2021	<i>Searches for new physics in Extended Higgs Sectors in CMS Phenomenology 2021 Symposium</i>
2020	<i>Precision Timing with Low Gain Avalanche Detectors for the CMS MTD Endcap Timing Layer [Poster]</i> IEEE NSS MIC Conference
2019	<i>Characterization of Hamamatsu LGADs for the CMS endcap timing layer and discussion of the test beam results [Poster]</i> Meeting of APS the Division of Particle and Fields
2019	<i>Test beam characterization of Hamamatsu LGADs for the CMS endcap timing layer [Poster]</i> Fermi National Laboratory 52nd Annual Users Meeting
2019	<i>Calibration and Alignment of the CMS Electromagnetic Calorimeter</i> APS April Meeting
2018	<i>CMS Electromagnetic Calorimeter and Alignment in LHC Run 2 [Poster]</i> 14th Pisa Meeting on Advanced Detectors
2018	<i>Performance of the CMS Electromagnetic Calorimeter data acquisition system at LHC Run 2 [Poster]</i> 14th Pisa Meeting on Advanced Detectors

SEMINARS

2022	<i>Development of the ATLAS Pixel Detector for the HL-LHC</i> Argonne National Laboratory, Young Scientists Symposium
2020	<i>Exotic decays of the Higgs with photons with the CMS detector</i> University of California Santa Barbara, Special HEP Seminar

AWARDS

2020	Dissertation Completion Fellowship Awarded final semester of funding to complete dissertation
2019	APS DPF Poster Session Award For poster presentation on “Characterization of Hamamatsu LGADs for the CMS endcap timing layer and discussion of the test beam result”
2019–2020	University Research Association Fellowship For developing precision timing upgrade of the CMS detector at Fermi National Laboratory
2017	Northeastern University Lawrence Award For excellence in teaching
2012–2014	Indira Gandhi Scholarship Awarded for excellence in M.Sc. Physics

SERVICE & OUTREACH

2024	Presented a talk on physics and ML, and answered questions in an interview by high school students, <i>Quarknet</i>
2023	Career panel and DEI session co-organizer, <i>US ATLAS Workshop</i>
2023	Chair of the Beyond Standard Model Physics session, <i>APS April DPF Meeting</i>
2022	Reviewer, <i>US DOE Science Graduate Student Research (SCGSR) Program</i>
2022	Committee member and symposium co-organizer, <i>Young Scientists Symposium at Argonne National Laboratory</i>
2021	Facilitator, <i>Northeastern Women in Physics Alumnae Panel [YouTube Link]</i>
2020	Exercise developer and instructor, <i>LPC Data Analysis School</i>

MENTORING

2023	Aonan Wang (PhD student, University of Science and Technology of China), ATLAS Run 3 WWW
2023	Xingyu Wu (PhD student, University of Science and Technology of China), ATLAS Run 3 WWW

2023	Runxuan Wang (PhD student, University of Michigan), ATLAS Run 3 WWW
2023	Daniel Aceves (Undergraduate student, University of Illinois, Chicago), ATLAS ITk Pixel Upgrade
2023	Jon Dale Nichols (PhD student, University of Oklahoma), ATLAS ITk Pixel Upgrade
2023	Matias Mantiñan (PhD student, University of Chicago), ATLAS ITk Pixel Upgrade
2023	Tate Flicker (Undergraduate student, University of Chicago), ATLAS ITk Pixel Upgrade
2023	Tanvi Rao (Undergraduate student, University of Chicago), Anomalous Quartic Gauge Couplings
2022	Daniel Perez (Undergraduate student, Susquehanna University), ATLAS ITk Pixel Upgrade
2022	Calvin Ainsworth (PhD student, Oklahoma State University), ATLAS ITk Pixel Upgrade
2020	Amrutha Krishna (PhD student, Northeastern University), CMS ECAL Operations

TEACHING

2015–2017	Northeastern University – Teaching Assistant for Undergraduate Courses • Physics for Life Sciences and Physics for Engineering: Laboratory instructor • Electricity and Magnetism: Tutored and graded quizzes and exams • Classical Mechanics: Taught recitation class for Physics for Engineering students
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